



Date: 23-10-2024

DEPARTMENT OF MATHEMATICS  
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

(An Institution of National Importance under Ministry of Education, Govt. of India)

MID SEMESTER EXAMINATIONS-II

B. Tech I Semester

MAIC-101: Differential Calculus and Differential Equations

Time: 09:30 AM to 10:20AM

Max. Marks: 20

N.B.: Answer ALL questions.

Answers to all parts of each question should be in one place.

|    |                                                                                                                                                                                                                               |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | Solve $\frac{d^4 t}{dy^4} + 18 \frac{d^2 t}{dy^2} + 81t = 0$                                                                                                                                                                  |
| Q2 | Determine functional dependence by Jacobian and find the relation between $u = \frac{x+y}{x-y}$ and $v = \frac{xy}{(x-y)^2}$ .                                                                                                |
| Q3 | The sum of three positive numbers is constant. Prove that their product is maximum when they are equal. (Hint: Use Lagrange's Method)                                                                                         |
| Q4 | Compute $f_{xy}(0,0)$ and $f_{yx}(0,0)$ for the function $f(x,y) = \begin{cases} \frac{xy^3}{x+y^2} : (x,y) \neq (0,0) \\ 0 : (x,y) = (0,0) \end{cases}$ . Also, discuss the continuity of $f_{xy}$ and $f_{yx}$ at $(0,0)$ . |

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